

ADMINISTRATION OF INTRANASAL FENTANYL FOR ACUTE PAIN IN THE CHILDREN'S EMERGENCY

1. BACKGROUND

- Intranasal (IN) fentanyl is a well-tolerated, safe, and effective method of pain management without the need for intravenous access or intramuscular injections.
- It has a bioavailability of 70 – 89%, onset of action of 2 minutes, duration of action of 30 – 60 minutes.
- There are no reported cases of opiate toxicity or death with IN fentanyl. (1)

1.1 INDICATIONS

- Fractures / MSK injuries (e.g., crush injuries)
- Burns

1.2 CONTRAINDICATIONS

- Allergic reaction or known hypersensitivity reaction to fentanyl.
- Mucosal abnormalities (mucus, blood and vasoconstrictors may reduce absorption)
- Altered mental state (GCS $\leq$ 14)
- Hemodynamic instability
- MAOI antidepressant use within the last 14 days (severe unpredictable potentiation of adverse effects)
- Age < 1 year (Relative contraindication – atomizer device unable to administer smaller doses effectively)

1.3 ADVERSE EFFECTS (2) (3) (4)

- Nausea, vomiting
- Dizziness
- Mild mucosal irritation
- Transient drowsiness, hypotension/hypoxia (no intervention required, spontaneous recovery)

1.4 PRECAUTIONS

Patients with renal and hepatic disease.

- No dosage adjustments are provided in manufacturer's labeling for patients with renal & hepatic disease, though pharmacokinetics may be altered. Consider use of IV analgesia (with established dose adjustments) vs. IN fentanyl (uncertain pharmacokinetics). Discuss with pharmacist/primary consultant if necessary.

2. DRUG PREPARATION & ADMINISTRATION

IN fentanyl should be administered by staff trained in its use.

2.1 DRUG PREPARATION

- Fentanyl 100mcg/2mL

2.2 DRUG DOSE

- 1st Dose = **1.5 mcg/kg**
- 2nd Dose = **0.75 to 1.5 mcg/kg**
- A second dose may be administered 5 to 10 minutes after the first to provide adequate analgesia. (Max. total dose = **3 mcg/kg**)

2.3 DRUG DELIVERY DEVICE

- The Teleflex® Mucosal Atomizer Device (MAD) Nasal™ device comes with 1mL and 3mL syringes. The device has a dead space volume of **0.1 mL**.
- For the 1st dose (**unprimed** MAD), an additional 0.1 mL should be drawn to account for the dead space. Subsequent doses with the same MAD do not require the additional 0.1 mL as the MAD has already been primed.

2.4 ORDERING IN FENTANYL

- The order for IN fentanyl should reflect the **intended dose** for administration (in mcg)
- Under admin. instructions (1st dose/unprimed MAD): **“Add 0.1ml for dead space.”**

2.5 DRUG ADMINISTRATION

1. Ensure patient is a suitable candidate for IN fentanyl with no drug allergies.
2. Perform suctioning of the nostrils (in an older patient, to blow their nose)
3. With a needle attached to the syringe, draw the correct volume of medication (add 0.1mL for the 1st dose)
4. Remove needle & attach MAD Nasal device to the syringe via the luer lock connector.
5. The patient should be seated with head extended at a 45° angle.
6. Place the tip of the MAD snugly against nostril aiming slightly upwards and outwards (toward the top of the ipsilateral ear) to cover the nasal turbinate and olfactory mucosa.
7. Briskly compress the syringe plunger to deliver half the medication into the nostril.
8. The dose should be divided between both nostrils.
9. The maximum volume per nostril is 1mL.
10. Do not discard the MAD Nasal device as it can be used for subsequent doses for the same patient. Label the device with the patient's sticky label for identification.
11. Subsequent doses can be administered 5 to 10 minutes later.

3. MANAGEMENT OF ADVERSE EVENTS

Mainly respiratory depression with hypoventilation.

1. Discontinue administration of opioid.
2. Stabilize Airway/Breathing/Circulation
3. Antidote = Naloxone

Indication	Naloxone Dose (IV/IM)
Hypoventilation (Spontaneously breathing)	5 mcg/kg/dose
Apnea/Impending respiratory arrest	< 5 years: 0.1 mg/kg/dose ≥ 5 years: 2mg
Dose can be given every 2 minutes based on response. (5) (Therapeutic endpoint for naloxone is adequate ventilation, not a normal level of consciousness.)	

4. RECOMMENDED DRUG DOSAGE FOR IN FENTANYL

Weight (kg)	Initial Dose (mcg) (1.5 mcg/kg)	Repeat Dose (mcg) (0.75 – 1.5 mcg/kg)
10	15	7.5 – 15
12	17.5	9 – 17.5
14	20	10 – 20
16	25	12.5 – 25
18	27.5	14 – 27.5
20	30	15 – 30
21 – 24	32.5	17.5 – 32.5
25	37.5	20 – 37.5
26 – 29	40	20 – 40
30 – 32	45	25 – 45
33 – 35	50	25 – 50
36 – 39	55	30 – 55
40 – 42	60	30 – 60
43 – 45	65	35 – 65
46 – 49	70	35 – 70
≥ 50	75	40 – 75

5. REFERENCES

1. *Intranasal fentanyl for the management of acute pain in children.* **Murphy A, O'Sullivan R, Wakai A, Grant TS, Barrett MJ, Cronin J, McCoy SC, Hom J, Kandamany N.** s.l. : Cochrane Database of Systematic Reviews, 2014.
2. *Intranasal analgesia for acute moderate to severe pain in children - a systematic review and meta-analysis.* **Prescott MG, Iakovleva E, Simpson MR, Pedersen SA, Munblit D, Vallersnes OM, Austad B.** 2023, BMC Pediatr.
3. *Equivalency of two concentrations of fentanyl administered by the intranasal route for acute analgesia in children in a paediatric emergency department: A randomized controlled trial.* **Borland, M., Milsom, S. and Esson, A.** s.l. : Emerg. Med. Australas, 2011.
4. *Intranasal fentanyl for initial treatment of vaso-occlusive crisis in sickle cell disease.* **Fein, D.M., et al.** s.l. : Pediatr. Blood Cancer, 2017.
5. **Naloxone: Pediatric Drug Information.** *UpToDate.* [Online] [Cited: March 21, 2024.] https://www.uptodate.com/contents/naloxone-pediatric-drug-information?search=naloxone&source=panel_search_result&selectedTitle=2%7E148&usage_type=panel&kp_tab=drug_pediatric&display_rank=1.
6. *Intranasal Fentanyl.* *The Royal Children's Hospital Melbourne.* [Online] https://www.rch.org.au/clinicalguide/guideline_index/intranasal_fentanyl/.